

Aliah University

End-Semester Examination - 2024

(For BCA. 2nd Semester)

Paper Name: Computer Organization

Paper Code: BCAUGMCC1202

Full Marks: 80

Time: 3 Hrs

Group – A

(10X1=10)

1. Fill in the blanks–

- CPU execution time = Instruction Count * _____ * Clock rate.
- Size of 1 cell in Byte addressable memories = _____.
- Computer data buses are _____ directional.
- Each division in cache memory (of fixed block size) is called _____.
- For a One-address instruction, any ALU operation is performed in _____.
- The following instruction - MOV 5(R1) is an example of _____ addressing mode.
- $(4AA1C)_{16}$ is equivalent to _____ in Octal.
- Cache Memory works on Principle of _____.
- Program Counter holds _____ of next instruction to be executed.
- 1-way set Associative mapping is exactly same as _____ mapping.

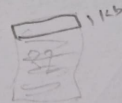
Group – B

Answer any six of the following questions -

(6X5=30)

- Explain the Little-endian and Big-endian concepts for Byte ordering with the help of an example. [5]
- Mention the names and the specific role of all special-purpose registers. [5]
- What is BUS? Explain various types of BUS in Computer Architecture. [1+4]
- What is "Hit ratio"? Draw a comparative discussion of various memory levels in computer memory hierarchy concept. (in short) [1+4]
- Define Addressing modes. Explain any 4-memory based addressing modes with examples.
- A Main memory of size 16 GB and a Cache Memory of size 32KB are divided into equal sized block and the Block size is 1KB. Mention number of bits require for performing – Direct mapping and 4-way/set associative mapping. [5]
- What is "overflow" in addition of 2's complemented numbers? Mention the conditions to check whether an overflow has occurred. Give examples to explain. [1+4]

$$16 \text{ GB} = 16 \times 2^{10} \times 2^{10} \text{ Kb}$$



$$\frac{16 \times 2^{20}}{32 \times 2^9}$$

$$\frac{16}{2^9}$$

Group – C

Answer any four of the following questions -

(10X4=40)

- What is Instruction cycle? Explain it with the help of a flowchart.
 - Consider a hypothetical system, which has 8-bit instruction and 2-bit addresses. How many 2-address instructions are possible? If there already exist 3 (Three) 2-address instruction and 5 (Five) 1-address instructions, then how many 0-address instructions are possible? [4+6]

$$16 - 3 - 5 = 8$$

3 2-ad
5 1-ad
0-ad

$$(2^2 - 3) - 5 = 1$$

$$48 + 2 + 2$$

10. a) Multiply (-11) and (-5) by using Booth's Algorithm.

b) Represent a decimal number $(-19)_{10}$ in 8-bit Sign-magnitude, 1's complement and 2's complement form. [7+3]

11. a) Suppose number is using 16-bit format: the 1-bit sign bit, 8-bits for signed exponent, and 7-bits for the fractional part. The leading bit 1 is not stored (as it is always 1 for a normalized number i.e. implicit normalization). Then represent $(-53.5)_{10}$.

b) What is the average memory access time for a machine with a cache hit rate of 80% and cache access time of 5 ns and main memory access time of 100 ns when Hierarchical access memory organization is used? [6+4]

$$(T_1 + T_2)(H_2)(1 - H_1)$$

12. a) What are the various Input/ Output transfer modes available? Write a brief note on DMA. *prog io, Interrupt*

b) Draw a comparative discussion between Hardwired, Vertical and Horizontal Designing concepts of Control Unit. [5+5]

13. Consider a system with 4 cache lines, with following main memory block references –
m m m m m m m m m m
9, 7, 53, 32, 9, 43, 18, 53, 9, 56, 34, 1, 32, 9

Assume initially cache is empty and then identify hit ratios by using

a) Direct Mapping Technique, *D*

b) 2-way set Associative with FIFO Approach. *$\frac{1}{13}$*

[4+6]

16	8	4	2	1
1	0	1	1	1
1	0	1	1	1